Aaron Besnoff

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EDUCATION

Lehigh University, Bethlehem, PA — Graduation: Spring 2027

Bachelor of Science in Mechanical Engineering; Minor in Aerospace Engineering

Advanced Coursework: Advanced Engineering Dynamics, Control Systems, Vibrations, Aircraft Design, Astrodynamics, Aerospace Engineering, Manufacturing, Fluid Mechanics, Thermodynamics

INDUSTRY EXPERIENCE

Engineering Intern – CEVA Logistics — June 2025 – August 2025

- Proposed and developed a **macro automation solution** that saved the company **500+ labor hours** in label creation and implementation.
- Assisted in planning, developing, and implementing a new **Warehouse Management System** (WMS).
- Designed a **Pick Pathing optimization system** to reduce inbound-to-outbound transfer times and improve workflow efficiency.

RESEARCH & DESIGN EXPERIENCE

Bomb Squad EOD X-Ray Deployment Robot – Drivetrain and Controls — January 2026 – Present

Lehigh University Senior Capstone, Bethlehem, PA

- Leading **chassis, suspension, and drivetrain design** for a low-cost teleoperated EOD robot built around portable X-ray payload, terrain, packaging, and cost requirements, reducing the welded 3/16 in stainless chassis by 44% (5.44 to 3.03 kg) using topology optimization and FEA at a 65 lb design load.
- Architecting a **Raspberry Pi and ESP32 control system** (Python, C++) linking an ELRS radio to four RS485 hub motors, with skid-steer and pivot modes, PI speed control with anti-windup, and firmware-enforced current, thermal, and stall protection, verified with nearly 800 automated tests and logged on-robot runs.
- Developed the **18 V lithium-ion power architecture**, including motor and battery selection, BMS integration, charging, power distribution, wiring, and protection, and defined failsafe behavior for radio loss, motor faults, emergency stopping, and deliberate system re-arming.

Bio-Inspired Multi-Medium UAV (Air/Water) – Lab Research — August 2025 – Present

Lehigh University, Moored Laboratory, Bethlehem, PA

- Collaborating with another undergraduate student to develop a **bio-inspired flapping-wing drone** capable of sustained operation in both aerial and underwater environments.
- Currently designing and prototyping a **waterproof thermal test stand** informed by principles of fluid dynamics, biomimicry, and aerospace design to evaluate water resistance and heat dissipation capabilities.
- Utilizing iterative design and testing to address challenges in waterproof sealing and propulsion efficiency across different mediums.

ONR MURI – Lab Research — January 2024 – August 2025

Lehigh University, Moored Laboratory, Bethlehem, PA

- Supported **ONR MURI** research on bio-inspired underwater propulsion and energy generation through development of a “Pitch and Heave” hydrofoil test rig.
- Independently designed a **custom belt-tensioning mechanism** in SolidWorks for the second-generation rig, creating fabrication drawings and integrating the mechanism into the assembly.
- Co-built the rig with another undergraduate researcher, including its aluminum structure and T-slot framing, using laser cutting, waterjet cutting, and mechanical assembly.

Founder – Lehigh University Rocketry Association — January 2024 – August 2025

- Developed initial design for a **liquid rocket motor thrust chamber**, and designed and built L1/L2 rockets using SolidWorks, OpenRocket, 3D printing, and machining.

Supersonic Aircraft Design Project – Aircraft Design — January 2025 – May 2025

- Designed and analyzed a supersonic air-superiority aircraft using **NASA OpenVSP/VSPAERO**, including aerodynamic configuration, sizing, and mission-performance analysis.

TECHNICAL SKILLS

- **Software/Programming: SolidWorks**, Python, **MATLAB**, Abaqus FEA, **OpenVSP**, **ANSYS STK**, Mastercam
- **Development Tools:** GitHub, Claude Code, Codex
- **Manufacturing:** CNC Milling, Lathe, Waterjet, 3D printing, Injection Molding